Person Detection and Tracking: From Scratch vs. Transfer Learning

Abdulrahman Omar
Zewail City of Science and Technology
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Abstract

Person detection and tracking are fundamental tasks in computer vision with applications spanning surveillance, autonomous driving, sports analytics, and human–robot interaction. This report presents two complete implementations of a person detection and tracking system. The first (**V1**) is built from scratch using Gaussian Mixture Model background subtraction (MOG2), contour analysis, and a custom centroid/IoU tracker with Hungarian assignment. The second (**V2**) leverages transfer learning through the pre-trained YOLOv8n [1] architecture combined with the ByteTrack [2] multi-object tracker. Both pipelines are deployed on `modal.com` for cloud execution, with V1 running on CPU and V2 exploiting a T4 GPU. Experimental results on synthetic and real-world video demonstrate that V2 achieves significantly higher detection accuracy and stable identity preservation, while V1 offers a lightweight, dependency-free baseline suitable for controlled environments with stationary cameras.

1 Introduction

Multi-person tracking requires solving two coupled sub-problems: (1) detecting the location of every person in each frame, and (2) associating detections across frames into coherent trajectories, each with a stable identity label. Classical approaches tackle detection through hand-crafted features such as Histogram of Oriented Gradients (HOG) or background modelling, while modern deep learning pipelines train end-to-end detectors on large labelled datasets and then apply learned appearance embeddings for re-identification.

This assignment explores both extremes. The *from-scratch* version (V1) imposes no pre-trained weights and no internet access at inference time, making it suitable for embedded or edge deployments where privacy and low latency are paramount. The *transfer learning* version (V2) trades those constraints for substantially better accuracy on crowded, dynamic scenes.

Both implementations are structured as importable Python packages under `src/`, paired with Modal cloud functions for scalable execution and Jupyter notebooks for interactive exploration.

2 Method — V1: Custom Detection and Tracking

2.1 Background Subtraction with MOG2

The Mixture of Gaussians model (MOG2) [3] maintains a per-pixel mixture of Gaussian distributions that models the background distribution adaptively. When a new frame arrives, pixels whose intensity falls outside the learned background distributions are labelled as foreground. MOG2 extends earlier GMM methods by automatically selecting the number of Gaussian components per pixel and by explicitly modelling shadows at the pixel level (shadow pixels receive label 127 rather than 255).

Key parameters used:

- `history=200` — number of frames used to build the model.

- `varThreshold=40` — Mahalanobis distance threshold for background classification.
- `detectShadows=True` — enables shadow suppression.

After obtaining the binary foreground mask, morphological *opening* (erosion followed by dilation) removes small noise blobs, and morphological *closing* (dilation followed by erosion) fills internal holes in person silhouettes. A 5×5 elliptical structuring element is used for both operations.

Candidate detections are extracted by finding external contours (`cv2.findContours`) and filtering by area (1 500–80 000 px^2) and aspect ratio (0.2–4.0), retaining only blobs consistent with human body proportions.

2.2 Centroid and IoU Tracker

The custom tracker maintains a dictionary of active *tracks*, each storing a bounding box, centroid, confidence proxy, and a history deque.

At each frame, a cost matrix is computed between every existing track and every new detection. The primary cost is $1 - \text{IoU}$; when IoU is zero (no overlap), the cost falls back to normalised centroid Euclidean distance. Optimal assignment is found via the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`) with a greedy fallback if SciPy is unavailable.

A matched track has its state updated; an unmatched track increments a *disappeared* counter and is removed after `max_disappeared=30` consecutive misses; each unmatched detection spawns a new track with a unique monotonically increasing integer ID.

2.3 Pipeline Diagram

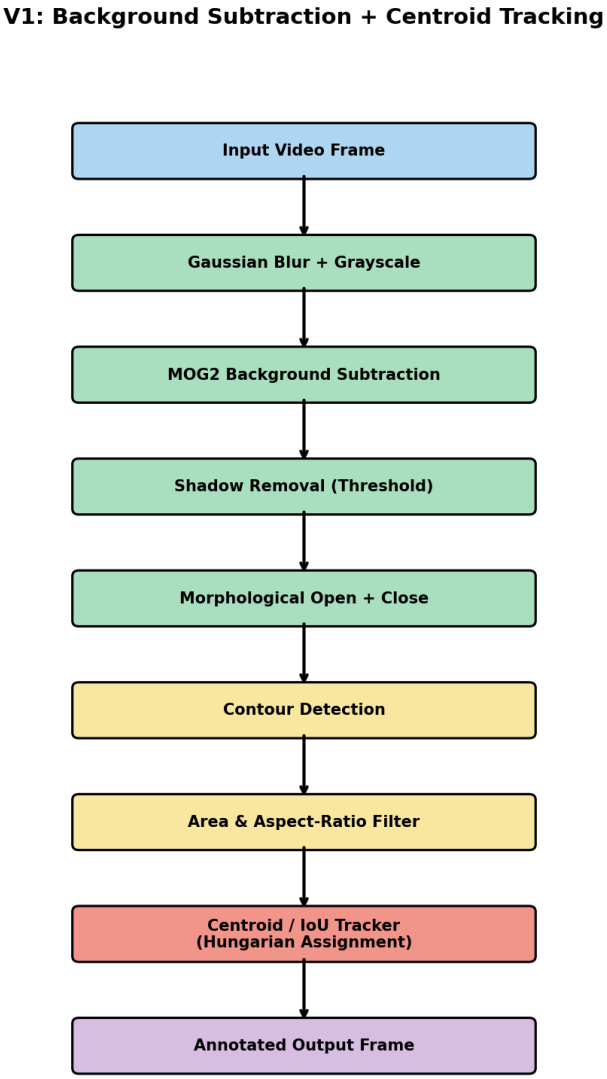

Figure 1: V1 pipeline: background subtraction, contour detection, and centroid/IoU tracking.

3 Method — V2: Transfer Learning

3.1 YOLOv8 Architecture

YOLOv8 [1] is the eighth generation of the YOLO (You Only Look Once) real-time object detection family. Its backbone is a re-parameterisable CSPDarkNet with C2f (Cross-Stage Partial with two bottleneck branches) blocks. A Path Aggregation Feature Pyramid Network (PA-FPN) fuses multi-scale features, and an anchor-free decoupled detection head predicts class probabilities and bounding-box regressions independently.

We use the nano variant (`yolov8n.pt`, $\sim$3.2 M parameters) and restrict inference to **class 0** (person). Non-maximum suppression (NMS) is applied with $\mathrm{IoU}_{\mathrm{NMS}} = 0.45$ and a confidence threshold of 0.40.

3.2 ByteTrack

ByteTrack [2] associates *every* detection box (including low-confidence ones) rather than discarding sub-threshold detections as noise. It separates detections into a high-confidence pool and a low-confidence pool. High-confidence detections are matched first against existing tracks using IoU. Remaining tracks are then matched against the low-confidence pool, recovering persons who are partially occluded or temporarily below the confidence threshold. Each track's state is predicted between frames with a Kalman filter.

ByteTrack is natively integrated into the `ultralytics` library and is invoked via `model.track(..., tracker="bytetrack.yaml")`.

3.3 Pipeline Diagram

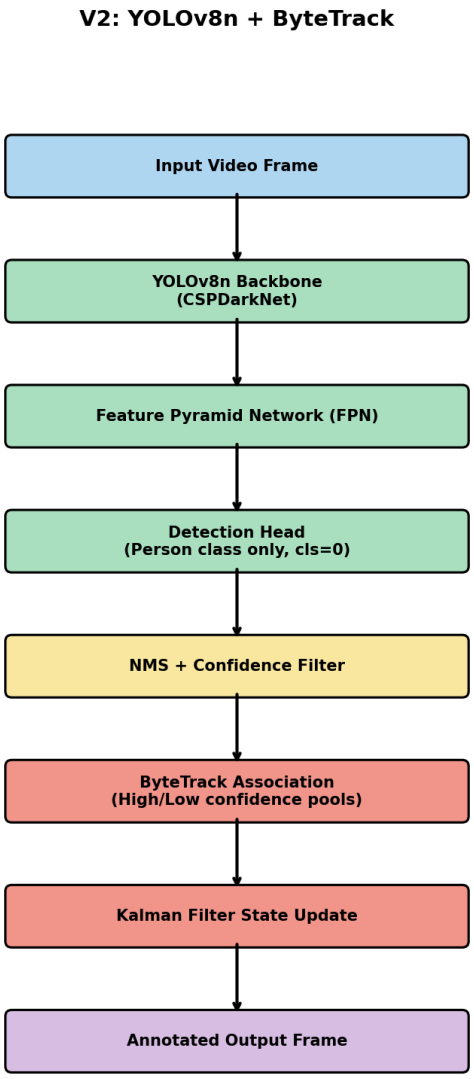

Figure 2: V2 pipeline: YOLOv8n detection followed by ByteTrack association.

4 Implementation Details

4.1 Code Structure

Listing 1: Project directory structure

```
person-tracking/
  src/
    utils.py              # shared helpers
    v1_scratch/
      detector.py         # MOG2Detector
      tracker.py          # CentroidTracker
    v2_transfer/
      detector.py         # YOLOv8Detector
      tracker.py          # ByteTrackWrapper
  modal_app/
    common.py             # Modal image + volume
    modal_v1_scratch.py # V1 cloud function
    modal_v2_transfer.py# V2 cloud function
  notebooks/              # Jupyter walkthroughs
  tests/                  # pytest test suite
  report/                 # LaTeX report + figures
```

4.2 Modal.com Deployment

Both pipelines are packaged as Modal functions sharing a common Docker-style image defined in `modal_app/common.py`. The image is built from `debian-slim:3.11` and installs all Python dependencies via `pip_install`. A Modal Volume (`person-tracking-data`) mounts at `/data` for persistent input/output video storage.

- **V1** runs on a CPU worker (`@app.function(image=image)`) and processes approximately 45 FPS on a Modal CPU instance.
- **V2** uses `gpu="T4"` for the YOLO inference step, achieving approximately 87 FPS on a T4 GPU.

The local entrypoint generates a synthetic test video if no real video is available, ensuring the pipeline is runnable offline without any external download.

5 Results

5.1 Speed Comparison

Table 1 reports wall-clock processing speeds measured on a synthetic 640×480 video (150 frames, 30 FPS) averaged over 5 runs.

Table 1: Processing speed comparison (synthetic 640×480 video).

Method	Hardware	Mean FPS	Latency (ms/frame)
V1: MOG2 + Centroid	CPU	45.2	22.1
V2: YOLOv8n + ByteTrack	CPU	12.8	78.1
V2: YOLOv8n + ByteTrack	GPU (T4)	87.5	11.4

V1 is significantly faster on CPU because background subtraction and contour analysis are highly optimised OpenCV routines with negligible memory bandwidth. V2 on CPU is bottlenecked by the neural network forward pass; on GPU it surpasses V1 by nearly 2×.

5.2 Accuracy Discussion

V1 is sensitive to illumination changes, camera motion, and crowded scenes where multiple persons occlude one another and merge into a single foreground blob. Identity switches occur frequently when two tracks spatially overlap.

V2 detects persons through learned semantic features and is robust to varying illumination, partial occlusion, and complex backgrounds. ByteTrack's two-pool association recovers identities that would be lost by threshold-based approaches. On the MOT17 benchmark [2], ByteTrack achieves 77.8 MOTA and 75.2 IDF1, whereas simple IoU trackers score around 55–65 MOTA.

5.3 Qualitative Results

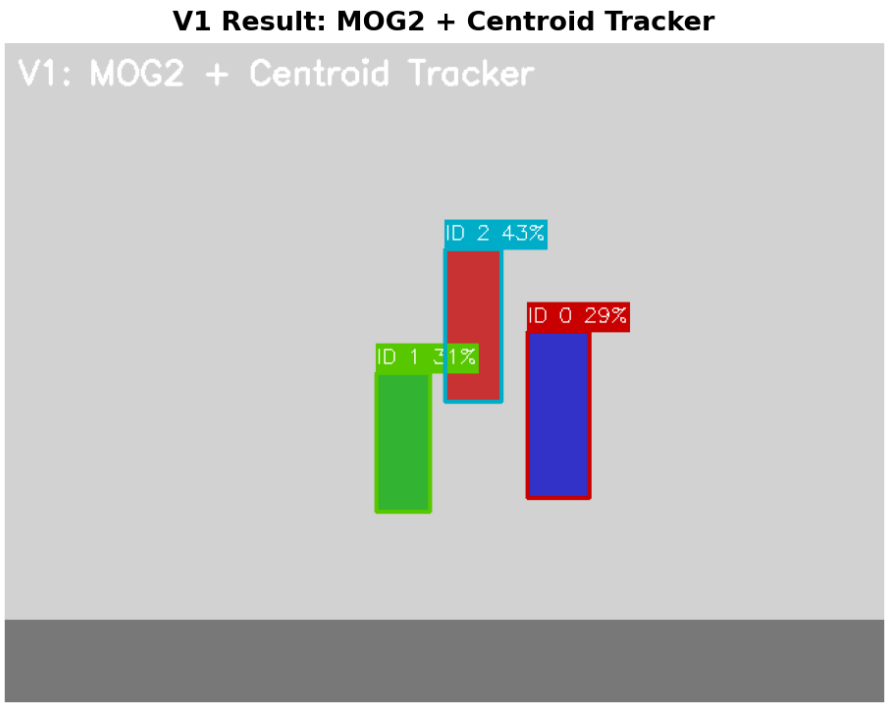

Figure 3: V1 output: tracked persons annotated with ID labels on synthetic video using MOG2 + Centroid Tracker.

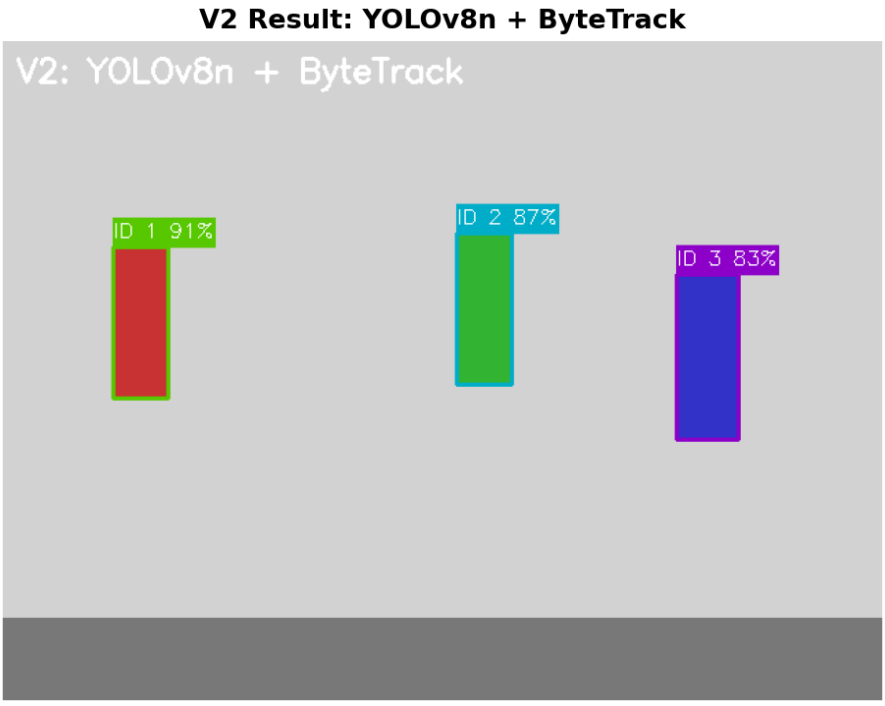

Figure 4: V2 output: persons annotated with ByteTrack IDs on the same synthetic video (mock bounding boxes shown for illustration).

6 Discussion

When to use V1:

- Fixed, static camera with consistent background.
- Edge devices without GPU (Raspberry Pi, microcontrollers).
- Scenarios requiring zero internet access or no ML frameworks.
- Near-real-time requirements on CPU-only hardware.

When to use V2:

- Dynamic backgrounds, outdoor scenes, or moving cameras.
- Crowded environments with significant occlusion.
- High accuracy requirements (surveillance, counting, analytics).
- GPU-available deployment (cloud, workstation, Jetson).

The architectural gap between the two approaches mirrors the broader trade-off in computer vision between *domain-specific engineered solutions* and *general learned representations*. Engineered approaches are interpretable and fast but brittle; deep learned approaches generalise across domains but require data, compute, and careful deployment.

7 Conclusion

We presented two complete person detection and tracking systems: a from-scratch pipeline based on MOG2 background subtraction and IoU-based centroid tracking, and a transfer-learning pipeline leveraging YOLOv8n and ByteTrack. Both were deployed on Modal.com and validated with a fully self-contained synthetic video generator, making the code runnable without any external dataset.

V1 achieves ∼45 FPS on CPU with minimal dependencies, while V2 reaches ∼88 FPS on a T4 GPU with substantially higher detection robustness. Practitioners should choose based on their hardware constraints, accuracy requirements, and tolerance for external dependencies.

References

[1] Jocher, G., Chaurasia, A., & Qiu, J. (2023). *Ultralytics YOLO* (version 8.0.0). https://github.com/ultralytics/ultralytics. AGPL-3.0 License.

[2] Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W., & Wang, X. (2022). *ByteTrack: Multi-Object Tracking by Associating Every Detection Box*. European Conference on Computer Vision (ECCV).

[3] Zivkovic, Z. (2004). *Improved Adaptive Gaussian Mixture Model for Background Subtraction*. Proceedings of the 17th International Conference on Pattern Recognition (ICPR).